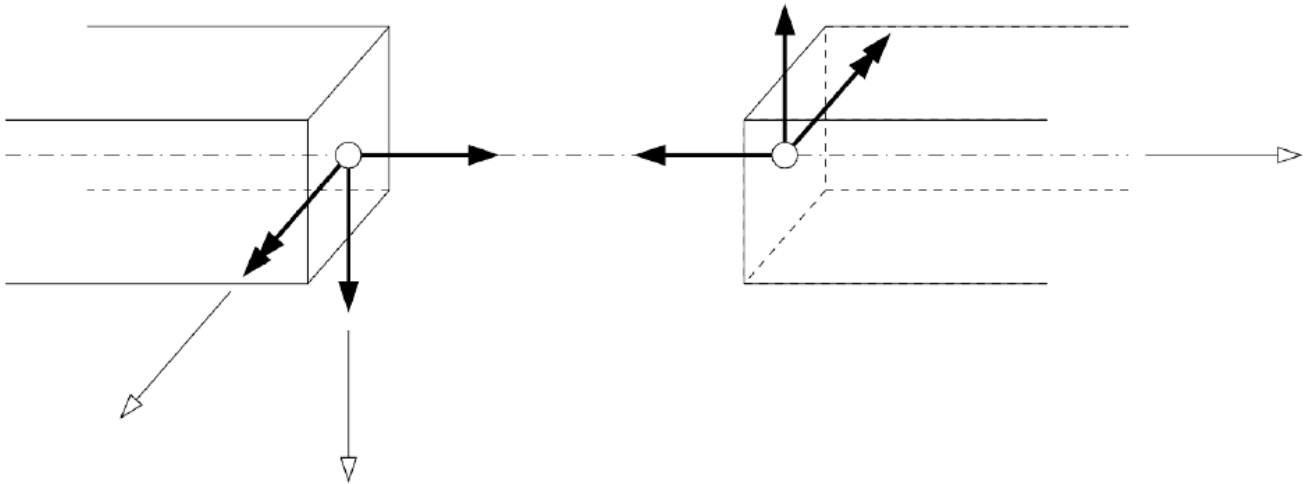
Engineering Mechanics I

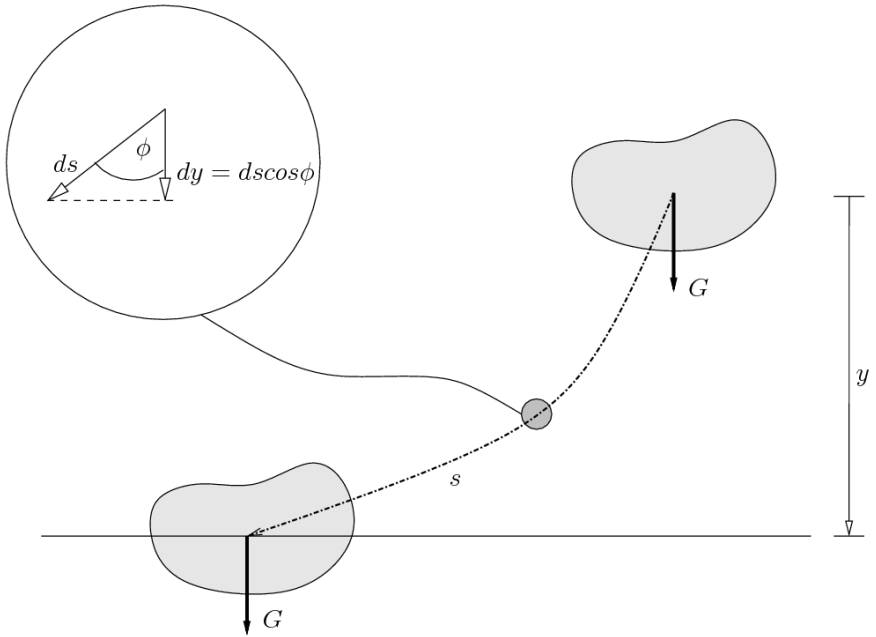
Lecture



Autonomous Aerial Systems

10.4.1. Conservative Forces

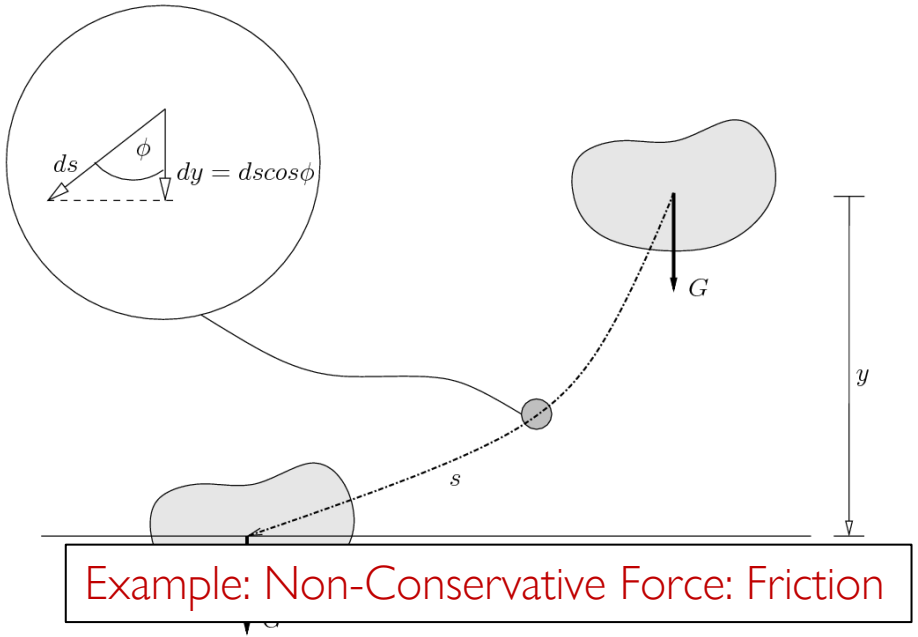
- Generated Work depends only on the starting and ending point, not on the course.
- Such forces are called conservative forces, example weight force:



Box 10.18

10.4.1. Conservative Forces

- Generated Work depends only on the starting and ending point, not on the course.
- Such forces are called conservative forces, example weight force:



Example: Non-Conservative Force: Friction

Box 10.18

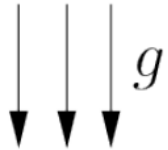
10.4.2. Potential Energy

- If a conservative force acts on a body, the body acquires the ability to perform work.
- One measure of this ability is potential or potential energy.
- Examples:
  - Weight force ↣ gravitational energy
  - Spring force ↣ elastic potential
  - Torsion spring moment ↣ elastic potential

10.4.2. Potential Energy - Gravitation potential

- **Gravitational force** loads all mass-bearing bodies with a force that depends on the **gravitational constant** g:

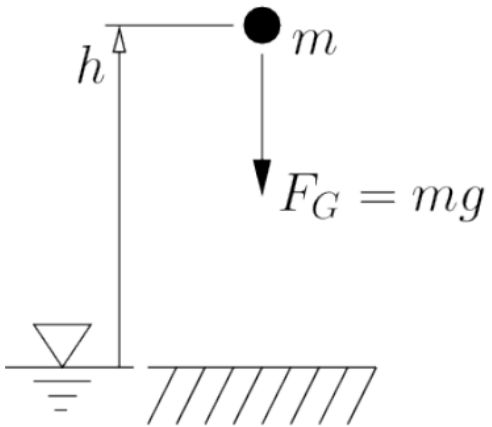
$$F_G = mg$$



- Direction of action of F_G along g.
- Potential of the gravitational force: **Gravitational potential/energy:**

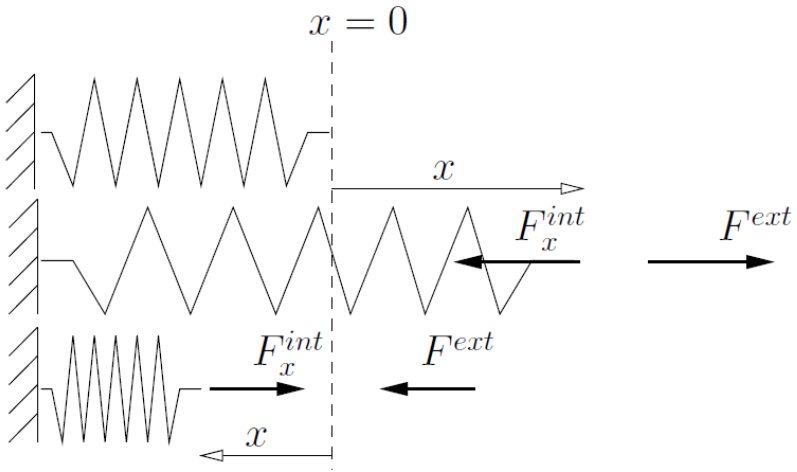
Box 10.19

- Height difference h measured against effective direction of g.



10.4.2. Potential Energy - Elastic potential

- Elastic potential is the potential energy stored in a body during deformation.
- Example: Spring under external load F^{ext}



Performed external work:

Box 10.20

- Internal work: It is performed by the elastic force (tension of the spring), storage as potential energy.

10.4.2. Potential Energy – Linear Elastic Spring

- Linear elastic spring -> force directly proportional to deformation
- Spring stiffness (spring constant): $k = \frac{F}{x} = const.$
- Work:

Box 10.21

- Elastic Potential Energy of a spring:

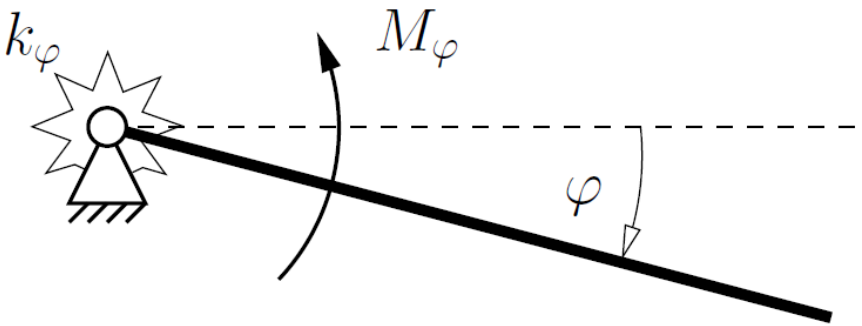
$$V_e = -W^{int} = \frac{1}{2}kx^2$$

- Spring Force via Differentiation:

$$F_{Feder}^{int} = -\frac{dV_e}{dx} = -kx$$

10.4.2. Potential Energy – Linear Elastic Rotational Spring

- Spring moment directly proportional to deformation
- Spring moment: $M_\varphi = -k_\varphi \cdot \varphi$
- Example:



  - Spring moment wants to turn back beam to initial position.
- Spring potential: $V_e = \frac{1}{2} k_\varphi \varphi^2$

10.4.2. Potential Energy – Potential of a Body

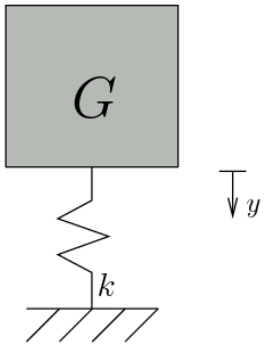
- The potential energy or the potential V of a body results from after definition of an initial position from the algebraic sum of gravitational energy and elastic energy:

$$V = V_g + V_e$$

- In general, the potential of a system depends on its n degrees of freedom q1,q2,...,qn summarized in the vector ,

$$V = V(\vec{q})$$

- Example single-mass oscillator with q = y:



Box 10.22

10.4.3. Potential as Equilibrium Condition

Consider a system with one degree of freedom q:

- Potential from infinitesimal displacement dq: $dV = -dW^{int}$
- Potential from virtual displacement δq: $\delta V = -\delta W^{int}$

At equilibrium, the virtual work must vanish. If the potential and its derivative are continuous, this condition is equivalent to:

Box 10.23

10.4.3. Potential as Equilibrium Condition

- For a system with n degrees of freedom, the total derivative (variation) must vanish:

$$V = V(q_1, q_2, \ldots, q_n)$$

$$\delta V = \frac{\partial V}{\partial q_1}\delta q_1 + \frac{\partial V}{\partial q_2}\delta q_2 + \ldots + \frac{\partial V}{\partial q_n}\delta q_n = 0$$

$\delta q_1, \delta q_2, \ldots, \delta q_n$ independent, therefore n equations

$$\frac{\partial V}{\partial q_1} = 0, \quad \frac{\partial V}{\partial q_2} = 0, \quad \ldots, \quad \frac{\partial V}{\partial q_n} = 0$$

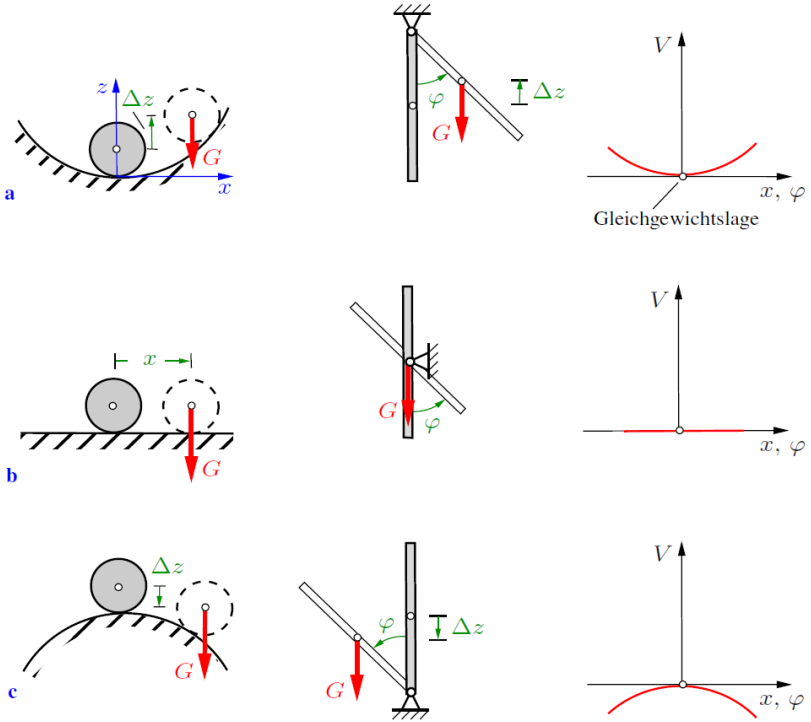
10.4.3 Stability in Equilibrium

Stability often plays an important role, e.g.

- Stable structure - static, dynamic
- Stability of an aircraft flight (aeroelasticity)
- Stability of a (numerical) solution method.

10.4.3 Stability in Equilibrium

- Conservative Systems with a single degree of freedom:



10.4.3 Stability in Equilibrium

Systematic consideration:

- **Necessary equilibrium condition** for system with one degree of freedom

$$\frac{dV(q)}{dq} = 0$$

- Equilibrium position is stationary point of potential function.
- Type of stationary point:

Box 10.24

Box 10.23

10.4.3 Stability in Equilibrium

- If $\frac{d^2V(q)}{dq^2} = 0$:higher derivatives decisive
  (i.e. positive or negative of a higher derivative))

- If $\frac{dV(q)}{dq} = 0 \wedge \frac{d^2V(q)}{dq^2} = \frac{d^3V}{dq^3} = \ldots = 0$ neutral equilibrium position,
  Potential is constant

- Systems with several degrees of freedom require more elaborate investigations, which are not considered here.